2. Quadratic Equations

1. Solve by factorisation: $x^2 - 7x + 12 = 0$.

2. Solve using quadratic formula: $3x^2 + 4x - 5 = 0$.

3. Solve: $2x^2 - 5x + 3 = 0$ and hence find the sum and product of its roots.

4. Find the discriminant of $5x^2 - 6x + 2 = 0$ and hence comment on the nature of its roots.

5. The product of two consecutive positive integers is 506. Form a quadratic equation and find the integers.

6. The sides of a rectangular park are in the ratio 3:2 and its area is 1500 m^2. Find the length and breadth of the park using a quadratic equation.

7. A pole is broken such that its top touches the ground and makes a triangle with the ground. The point of break is 5 m above the ground and the top of the pole touches the ground at a distance of 12 m from the foot of the pole. Find the original height of the pole.

8. If α and β are the roots of $x^2 - 4x + 3 = 0$, form the quadratic equation whose roots are 2α and 2β.

9. Solve for x: $1/x + 1/(x - 2) = 3/4$.

10. The sum of the reciprocals of two consecutive natural numbers is 11/30. Find the numbers by forming and solving a quadratic equation.